

Beyond Newton's equations: sampling statistical ensembles

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

Day 1/5

Dr. Vittorio Del Tatto

Statistical mechanics

- Classical mechanics: with N particles, the state of the system is defined by $6N$ numbers
- However in the real world, only few **macroscopic variables** can be typically observed
- If we measure a set of macroscopic variables, can we still say something about the state of the underlying **microscopic variables**?

Statistical mechanics studies the relationship between macroscopic quantities (e.g. temperature, pressure, specific heat) and the microscopic degrees of freedom



N.B. involve **ensemble averages** of the microscopic variables

- **Example:** relationship between average kinetic energy and temperature,

$$\langle K \rangle = \frac{3}{2} N k_B T \quad \text{“equipartition theorem”}$$

Macroscopic and microscopic states

- **Macroscopic state:** identified by the state of one or more macroscopic variables
 - e.g. ($N = 100$, $V = 10 \text{ nm}^3$, $T = 273.15 \text{ K}$)
- **Microscopic state:** identified by the state of all the underlying (unknown) microscopic variables ($3N$ positions, $3N$ momenta)
 - e.g. $\Gamma = (\mathbf{q}_1, \dots, \mathbf{q}_N, \mathbf{p}_1, \dots, \mathbf{p}_N)$

A large (often infinite) number of microscopic states can represent the same macroscopic state!

- **Statistical ensemble:**
 - a macroscopic state
 - the set of microscopic states compatible with the macroscopic state, $\{\Gamma\}$
 - the probability of finding the system in each microscopic state, $p(\Gamma)$

Properties of statistical ensembles

- **Equilibrium:**

- probability $\mathcal{P}(\Gamma)$ does not depend on time: $\frac{\partial \mathcal{P}(\Gamma)}{\partial t} = 0$
- probability $\mathcal{P}(\Gamma)$ is invariant under time-reversal symmetry:

$$\mathcal{P}(\Gamma) = \mathcal{P}(\Gamma^*) \text{ where } \Gamma^* = (\mathbf{q}_1, \dots, \mathbf{q}_N, -\mathbf{p}_1, \dots, -\mathbf{p}_N)$$

- **Ensemble averages:** given microscopic observable $O(\Gamma)$, its expected value is

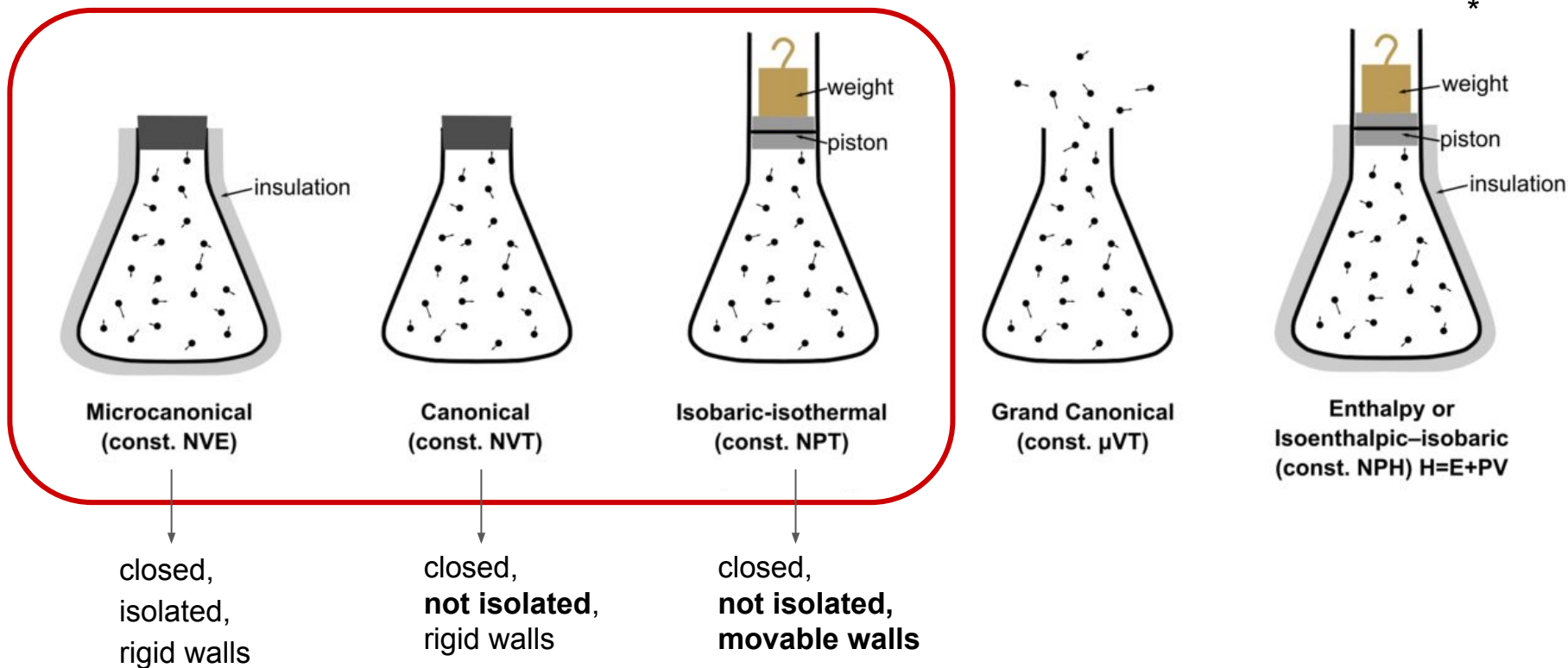
$$\langle O \rangle = \int d\Gamma O(\Gamma) \mathcal{P}(\Gamma)$$

- **Ergodicity:** if $\Gamma(t)$ denotes a **single trajectory** starting from $\Gamma(0)$,

$$\lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T dt O(\Gamma(t)) = \langle O \rangle$$

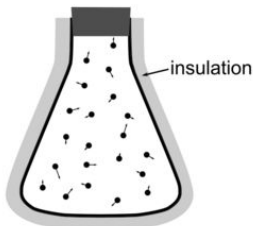
If the system is ergodic, ensemble averages (across different realizations) are equivalent to time averages (on a single realization)!

Statistical ensembles



Statistical ensembles

Microcanonical
(const. NVE)

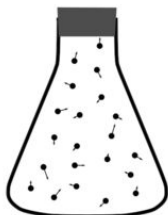


$$\Gamma = \{\mathbf{q}_i, \mathbf{p}_i\}$$

$$\mathcal{P}_{\text{NVE}}(\Gamma) = \frac{1}{\Omega_{\text{NVE}}} \delta(H(\Gamma) - E)$$

from principle of “equal a priori probability”

Canonical
(const. NVT)

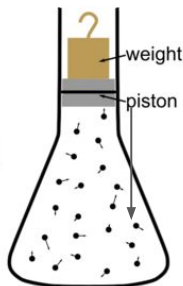


$$\Gamma = \{\mathbf{q}_i, \mathbf{p}_i\}$$

$$\mathcal{P}_{\text{NVT}}(\Gamma) = \frac{1}{Z_{\text{NVT}}} \exp(-\beta H(\Gamma))$$

Boltzmann distribution

Gibbs or
Isobaric-isothermal
(const. NPT)



$$\Gamma = (\{\mathbf{q}_i, \mathbf{p}_i\}, V)$$

$$\mathcal{P}_{\text{NPT}}(\{\mathbf{q}_i, \mathbf{p}_i\}, V) = \frac{1}{Z_{\text{NPT}}} \exp\left(-\beta(H(\{\mathbf{q}_i, \mathbf{p}_i\}) + pV)\right)$$

Isothermal-isobaric
distribution

$$\beta = \frac{1}{k_B T}$$

Canonical ensemble (NVT)

- Distribution we want to sample: $\mathcal{P}_{\text{NVT}}(\Gamma) \propto \exp(-\beta H(\Gamma))$
- **Problem:** direct sampling is not possible! ($\neq$ simple low-dimensional distribution)
- **Solution:** complex probability distribution can be still sampled using Markov chains!

A Markov-chain sampler of a distribution $\mathcal{P}_{\text{target}}(\Gamma)$ is an update rule $\mathcal{S} : \Gamma \mapsto \Gamma'$ which should satisfy the following properties:

1. **Stationarity of the target distribution:** applied to $\{\Gamma_i\} \sim \mathcal{P}_{\text{target}}$, it returns $\{\Gamma'_i\} \sim \mathcal{P}_{\text{target}}$
2. **Ergodicity:** no “unexplored” regions in the Γ - space after a finite number of updates
3. [optional] **Fast decorrelation of samples:** $\mathcal{P}_{\text{target}}(\Gamma)$ sampled with short trajectories!

N.B. the update rule can be **deterministic** or **stochastic** (involving the extraction of random numbers)

Canonical ensemble (NVT)

Candidate Markov chain samplers:

- Markov chain Monte-Carlo (not in this school): update in two steps,
 - deterministic **trial move**
 - random **acceptance rule** (Metropolis-Hastings)

Limitation: designing trial moves which give large acceptance is hard! (property 3 not satisfied)

- Velocity Verlet? Is it a good sampler of $\mathcal{P}_{\text{NVE}}(\Gamma)$?
 - deterministic update: $\Gamma = \{\mathbf{q}_i(t), \mathbf{p}_i(t)\} \rightarrow \Gamma' = \{\mathbf{q}_i(t + \Delta t), \mathbf{p}_i(t + \Delta t)\}$
 - Property 1 (stationarity)? satisfied (can be shown)
 - Property 2 (ergodicity)? not satisfied: dynamics stuck to subregion of phase space with

$$H(\Gamma) = \text{const} = H(\Gamma_0)$$

Canonical ensemble (NVT)

- Can we couple Velocity Verlet to an update rule that modifies energy according to the NVT distribution, achieving ergodicity?



If we could find an **energy update rule** which is stationary wrt NVT distribution, we would achieve ergodicity while preserving stationarity!

In other words, we would have a sampler of the canonical distribution!

Canonical ensemble (NVT)

- It turns out that a simple way to do this is **updating only the kinetic energy part**
- How? Distribution is separable in positions and momenta: since $H(\Gamma) = U(\{\mathbf{q}_i\}) + K(\{\mathbf{p}_i\})$,

$$\begin{aligned}\mathcal{P}_{\text{NVT}}(\{\mathbf{q}_i\}, \{\mathbf{p}_i\}) &\propto \exp(-\beta U(\{\mathbf{q}_i\})) \exp(-\beta K(\{\mathbf{p}_i\})) \\ &\propto \underbrace{\exp(-\beta U(\{\mathbf{q}_i\}))}_{\text{difficult to sample}} \underbrace{\prod_{i=1}^N \exp\left(-\beta \frac{\mathbf{p}_i^2}{2m_i}\right)}_{\substack{\text{easy to sample} \\ \text{(just a product of Gaussians!)}}}\end{aligned}$$

A **sampling algorithm for the kinetic energy** of the NVT distribution is called **thermostat**

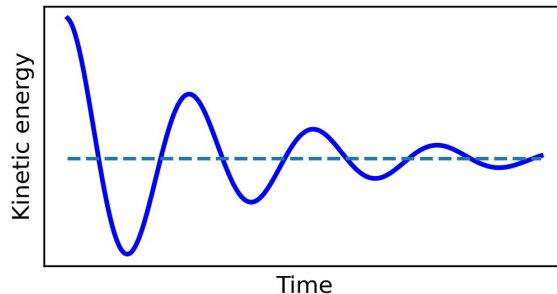
Hamilton's eqs. integrator (e.g. Velocity Verlet) + thermostat = sampler of canonical ensemble

Thermostats

Properties:

- **Deterministic** vs **stochastic** thermostats

Typically equilibrate faster (no slow damped oscillations), exception: Berendsen



- **Local** vs **global** thermostats

update momenta $\{\mathbf{p}_i\}$ one by one,
with a different rule per particle

requires extracting *3N random numbers*

update momenta $\{\mathbf{p}_i\}$ globally,
with the same rule

requires extracting *a single random number*

Thermostats

A list of thermostats:

- **Berendsen thermostat:** deterministic, global (but technically **wrong!** It samples correctly the mean kinetic energy, not the full distribution)
—————→ use it for **equilibration**, not for **production**
- **Andersen thermostat:** stochastic, local
- **Langevin thermostat:** stochastic, local
- **Nosé-Hoover:** deterministic, global
- **Stochastic velocity rescaling (v-rescale):** stochastic, global

Berendsen thermostat

- Idea:** relax exponentially the kinetic energy towards its average value in the NVT distribution

Equipartition theorem: $\overline{K} = \frac{3}{2} N k_B T$ ← What is this value?

P.s. Now we can also derive this quantity, from NVT distribution of the kinetic energy

Change of variable: $\mathcal{P}_{\text{NVT}}(K) = \int d\Gamma \delta(K - K(\Gamma)) \mathcal{P}_{\text{NVT}}(\Gamma)$

“Gamma
distribution”

$$\begin{array}{ccc} \downarrow [\dots] & & \\ \boxed{\mathcal{P}_{\text{NVT}}(K) \propto K^{\frac{N_f}{2}-1} \exp(-\beta K)} & \xrightarrow{[\dots]} & \boxed{\langle K \rangle = \frac{N_f}{2} k_B T} \end{array}$$

where $N_f =$ number of momentum degrees of freedom

Berendsen thermostat

- Exponential relaxation is the solution of this differential equation:

$$\frac{dK}{dt} = \frac{\overline{K} - K}{\tau} \longrightarrow \text{characteristic relaxation timescale}$$

- The solution is:
$$K(t + \Delta t) = \underbrace{e^{-(t+\Delta t)/\tau} K(t)}_{\substack{\text{Exponential suppression} \\ \text{of initial kinetic energy}}} + \left(1 - e^{-(t+\Delta t)/\tau}\right) \overline{K}$$
- In practice: an update at each MD step, then velocity Verlet, then update, ...

Berendsen thermostat

Algorithm 3 Berendsen thermostat

```
for  $n = 1, \dots, N_{\text{steps}}$  do
   $\{\mathbf{q}_i\}, \{\mathbf{p}_i\} \leftarrow \text{VELOCITYVERLET}(\Delta t)$ 
   $K \leftarrow \sum_i |\mathbf{p}_i|^2 / (2m_i)$ 
   $K' \leftarrow e^{-\Delta t/\tau} K + (1 - e^{-\Delta t/\tau}) \bar{K}$ 
   $\alpha \leftarrow \sqrt{K'/K}$ 
   $\mathbf{p}_i \leftarrow \alpha \mathbf{p}_i \quad \forall i$ 
end for
```

↘ this final scaling ensures that new momenta are
“compatible” with the new kinetic energy

- ✓ relaxation is fast and tuned by simple parameter
- ✗ wrong target distribution

Andersen thermostat

- Idea: **re-sample momenta from scratch every few steps** (using a stride n_s)

Algorithm 1 Andersen thermostat

```
for  $n = 1, \dots, N_{\text{steps}}$  do
   $\{q_i\}, \{p_i\} \leftarrow \text{VELOCITYVERLET}(\Delta t)$ 
  if  $n \bmod n_s = 0$  then
    for  $i = 1, \dots, N$  do
       $p_i \sim \mathcal{P}(p_i) \propto \exp\left(-\beta \frac{|p_i|^2}{2m_i}\right) \equiv \mathcal{N}(\mathbf{0}, m_i k_B T \mathbf{I}_3)$ 
    end for
  end if
end for
```

- ✓ samples the canonical distribution
- ✓ ✗ equilibration of momenta is instantaneous, equilibration of positions can be slow

Langevin thermostat

- can be seen as a “softer” version of the Andersen thermostat

Algorithm 2 Langevin thermostat

```
for  $n = 1, \dots, N_{\text{steps}}$  do
   $\{q_i\}, \{p_i\} \leftarrow \text{VELOCITYVERLET}(\Delta t)$ 
  for  $i = 1, \dots, N$  do
     $\xi_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_3)$ 
     $p_i \leftarrow e^{-\Delta t/\tau} p_i + \sqrt{(1 - e^{-2\Delta t/\tau}) m_i k_B T} \xi_i$ 
  end for
end for
```

- ✓ samples the canonical distribution
- ✓✗ it's a local thermostat: achieves the right distribution by perturbing many degrees of freedom independently (not a small perturbation) → relaxation time affected

Stochastic velocity rescaling

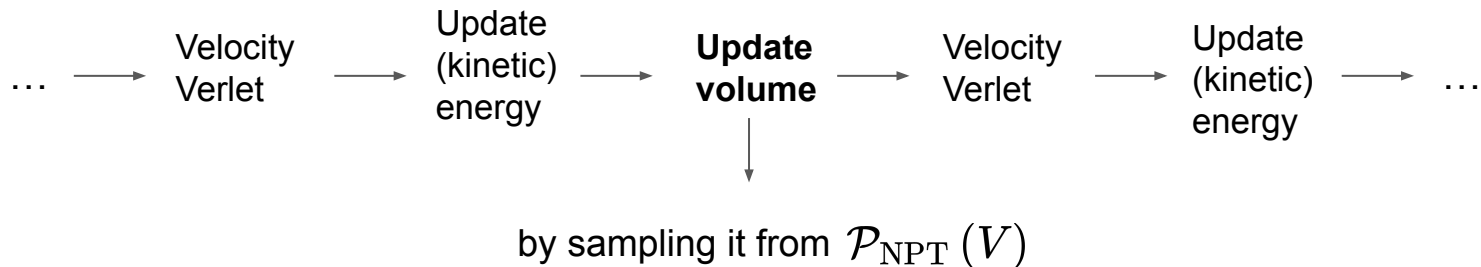
- can be seen a stochastic version of the Berendsen barostat which targets the correct distribution

Algorithm 4 Stochastic velocity rescaling (SVR)

```
for  $n = 1, \dots, N_{\text{steps}}$  do  
   $\{\mathbf{q}_i\}, \{\mathbf{p}_i\} \leftarrow \text{VELOCITYVERLET}(\Delta t)$   
   $K \leftarrow \sum_i |\mathbf{p}_i|^2 / (2m_i)$   
   $R_1 \sim \mathcal{N}(0, 1), \quad S \sim \chi^2_{N_f - 1}$   
   $K' \leftarrow e^{-\Delta t/\tau} K + \frac{\bar{K}}{N_f} \left(1 - e^{-\Delta t/\tau}\right) (R_1^2 + S)$   
   $\quad + 2R_1 \sqrt{\frac{K \bar{K}}{N_f} e^{-\Delta t/\tau} (1 - e^{-\Delta t/\tau})}$   
   $\alpha \leftarrow \sqrt{K'/K}$   
   $\mathbf{p}_i \leftarrow \alpha \mathbf{p}_i \quad \forall i$   
end for
```

- ✓ samples the canonical distribution
- ✓ it's a global thermostat: fast equilibration

Isothermal – isobaric ensemble (NPT)



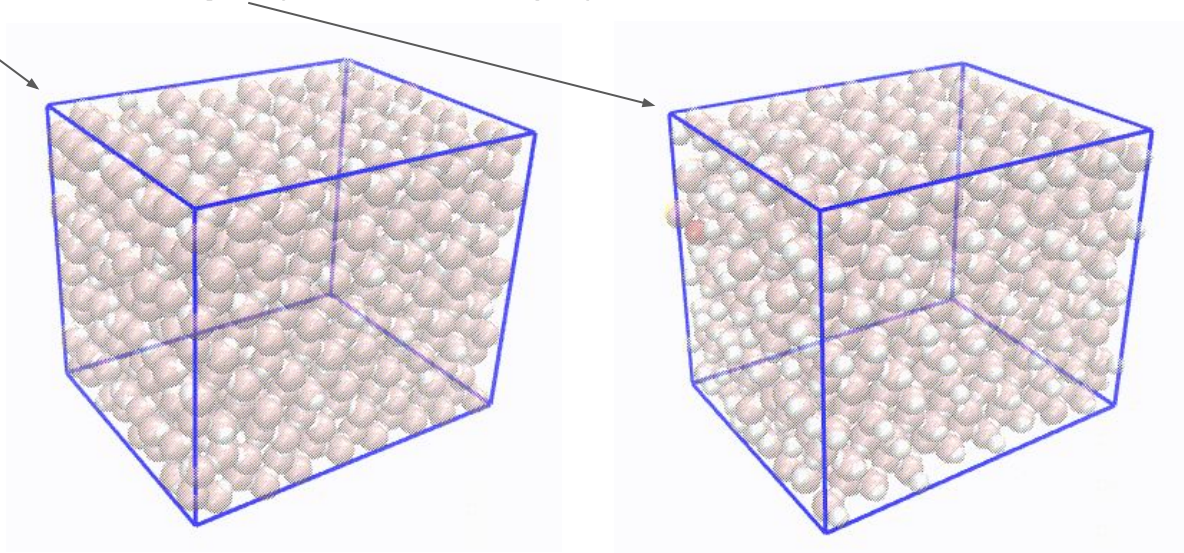
A **sampling algorithm for the volume** of the NPT distribution is called **barostat**

Hamilton's eqs. integrator + thermostat + barostat = sampler of NPT ensemble

Barostats

Properties:

- **Deterministic** vs **stochastic** barostats
- ~~Local vs global?~~
- **Isotropic** vs **anisotropic** (vs semi-isotropic)



Barostats

A list of barostats:

- **Berendsen:** deterministic
- **Parrinello-Rahman:** deterministic
- **Martyna-Tuckerman-Tobias-Klein (MTTK):** deterministic
- **Stochastic cell rescaling (c-rescale):** stochastic

Backup slides

Canonical ensemble (NVT)

- Distribution is separable in positions and momenta: since $H(\Gamma) = U(\{\mathbf{q}_i\}) + K(\{\mathbf{p}_i\})$,

$$\mathcal{P}_{\text{NVT}}(\{\mathbf{q}_i\}, \{\mathbf{p}_i\}) \propto \exp(-\beta U(\{\mathbf{q}_i\})) \exp(-\beta K(\{\mathbf{p}_i\}))$$

- Canonical distribution as a function of the kinetic energy only?

Change of variable: $\mathcal{P}_{\text{NVT}}(K) = \int d\Gamma \delta(K - K(\Gamma)) \mathcal{P}_{\text{NVT}}(\Gamma)$

[...]

$$\mathcal{P}_{\text{NVT}}(K) \propto K^{\frac{N_f}{2}-1} \exp(-\beta K)$$

“Gamma distribution”

where $N_f = 3N - N_c - 3$

number of momentum degrees
of freedom

momentum
components

number of constraints
(e.g. rigid bonds, fixed
angles)

number of conservation
rules (if no external field)

Isothermal – isobaric ensemble (NPT)

- Target distribution:

$$P_{\text{NPT}}(\{\mathbf{q}_i, \mathbf{p}_i\}, V) \propto \exp(-\beta(H(\{\mathbf{q}_i, \mathbf{p}_i\}) + pV))$$

$$\propto \frac{Z_{\text{NVT}}}{Z_{\text{NVT}}} \exp(-\beta H(\{\mathbf{q}_i, \mathbf{p}_i\})) \exp(-\beta pV)$$

$$\propto \underbrace{\mathcal{P}_{\text{NVT}}(\{\mathbf{q}_i, \mathbf{p}_i\})}_{\text{now we know how to sample it}} \underbrace{Z_{\text{NVT}} \exp(-\beta pV)}_{\text{what we need to sample! } \mathcal{P}_{\text{NPT}}(V)}$$

(velocity Verlet + thermostat)

- N.B. $\{\mathbf{q}_i, \mathbf{p}_i\}$ and V are separated in the last expression, so:
 - a sampler of the NVT distribution is **stationary** wrt NPT distribution...
 - ...but **not ergodic** (volume fixed)!

does it ring a bell?

Canonical ensemble (NVT)

- Distribution is separable in positions and momenta: since $H(\Gamma) = U(\{\mathbf{q}_i\}) + K(\{\mathbf{p}_i\})$,

$$\mathcal{P}_{\text{NVT}}(\{\mathbf{q}_i\}, \{\mathbf{p}_i\}) \propto \exp(-\beta U(\{\mathbf{q}_i\})) \exp(-\beta K(\{\mathbf{p}_i\}))$$

- Canonical distribution as a function of the kinetic energy only?

Change of variable: $\mathcal{P}_{\text{NVT}}(K) = \int d\Gamma \delta(K - K(\Gamma)) \mathcal{P}_{\text{NVT}}(\Gamma)$

[...]

$$\mathcal{P}_{\text{NVT}}(K) \propto K^{\frac{N_f}{2}-1} \exp(-\beta K)$$

“Gamma distribution”

where $N_f = 3N - N_c - 3$

number of momentum degrees of freedom

number of momentum components

number of constraints (e.g. rigid bonds, fixed angles)

number of conservation rules (if no external field)

Deriving the Boltzmann distribution

How to derive it?

1. From **microcanonical ensemble**, splitting a single NVE system in two NVT systems (reservoir + system)
2. From an information-theoretic principle: **Maximum Entropy (MaxEnt)**



For simplicity we will write it for a discrete distribution:

$$\left. \begin{array}{l} \Gamma \in \mathbb{R}^{6N} \rightarrow \Gamma \in \{\Gamma_i\} \\ p(\Gamma) \rightarrow p_i = p(\Gamma_i) \end{array} \right\} \begin{array}{l} \text{error of the approximation can be made arbitrarily small} \\ \text{by taking a grid which is “fine enough”} \end{array}$$

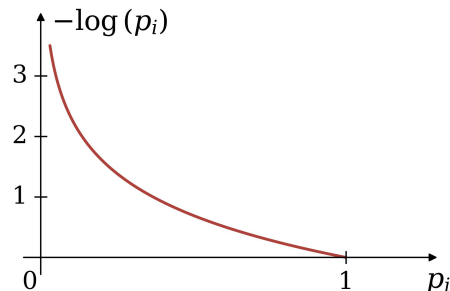
[Just a matter of convenience: same results can be obtained in the continuous case]

Deriving the Boltzmann distribution

Definitions (from Information Theory)

- **Surprisal (of an event):** *“How unlikely is the configuration Γ_i ?”*

$$s(\Gamma_i) = -\log(p_i)$$



- **Shannon entropy (of the distribution):** *“How unlikely is a generic configuration $\Gamma \in \{\Gamma_i\}$?”*

$$S[p] = \mathbb{E}[s(\Gamma)] = -\sum_i p_i \log p_i$$